BIG DATA ENGINEERING · 2025

Turkey Food Supply Chain Transparency & Margin Analysis

A Real-Time Decision Support Engine for Turkey's Agricultural Value Chain

THE PROBLEM

Tomato enters Istanbul's Hall at ₺10 — hits the retail shelf at ₺35. *Nobody can explain the gap.*

OUR ANSWER

A real-time pipeline ingesting 8 independent sources, tracking every margin step across 81 provinces.

BUILT FOR

Decision Support System for Turkey's Ministry of Trade, Competition Authority & regulators.

Apache Kafka · Apache Flink · Delta Lake · Airflow · Superset · Medallion Architecture

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PROBLEM STATEMENT

Turkey's Food Pricing is a Black Box

₺10
Wholesale Hall Entry
(İBB Hal)

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THE BLACK BOX
+₺25 unaccounted

→

₺35
Retail Shelf
(Kadıköy Market)

Systematic Data Void

No unified, machine-readable pipeline tracks the farm-to-shelf journey. The supply chain is a **completely unmeasured black box** — regulators and citizens have no real-time data to act on.

Asymmetric Price Pass-Through ("Rockets & Feathers")

When input costs rise, retail prices **rocket up instantly**. When costs fall, prices **feather down slowly** — or not at all. This asymmetry is a structural consumer harm never proven with data.

Location-Based Speculation

Same wholesale entry price → drastically different district-level retail prices. **Istanbul Kadıköy: ₺35** vs **Izmir Karşıyaka: ₺25** for the same product from the same hall. No accountability mechanism exists.

Unmeasured Supply Shock Propagation

When a frost hits Antalya, **how many days** until Istanbul's retail shelves reflect the shock? When energy prices rise, **how much** of the logistics cost increase passes through to the consumer? These chains are invisible.

GOAL & SOLUTION

From Opacity to a National Intelligence Layer

> *Build a real-time **Decision Support System (DSS)** and **Early Warning Radar** for Turkey's Ministry of Trade, Competition Authority, and market regulators — powered by 8 independent live data streams.*

FOUR RESEARCH QUESTIONS WE ANSWER

- **Spatial Margin Analysis** — Province & market-level hall-to-shelf margin decomposition
- **Shock Propagation Speed** — Days from weather event → wholesale → retail
- **Rockets & Feathers** — Asymmetric price pass-through quantified with lag analysis
- **Logistics Cost Driver** — Energy/FX correlation with margin widening across provinces

WHAT WE BUILD

- Real-time ingestion of **8 independent data sources** in 3 different formats
- **Entity resolution engine** merging hall product names with retail SKUs
- **Medallion architecture** on Delta Lake (Bronze → Silver → Gold)
- **Superset dashboards**: Turkey Margin Map, Rockets & Feathers Radar, Shock Event Overlay

8	50–500GB	81	15min	50K+
DATA SOURCES	SCALABLE VOLUME	PROVINCES	FASTEST CADENCE	SKUS TRACKED

DATA SOURCES

8 Independent Streams — Unified Pipeline

SOURCE	PROVIDER	FORMAT	VOLUME	CADENCE	INGESTION
Retail Prices	marketfiyati.org.tr	REST JSON	15–20 GB/mo	Intraday	Kafka Stream
IBB Wholesale Hall	IBB Open Data (Swagger)	REST JSON	2–3 GB/yr	Daily	Airflow Batch
Izmir Hall Prices	Izmir BB Open Data	CSV / JSON	1 GB/yr	Daily	Airflow Batch
IBB Hall Tonnage	IBB Open Data (Bulk)	CSV	500 MB hist.	Monthly	Airflow Batch
Weather (Agricultural)	Open-Meteo API	JSON	5 GB/yr	Hourly	Kafka Stream
Food & Agri News	GDELT Project (S3)	CSV.gz	10–15 GB/yr	15-min	Kafka Stream
Electricity Prices	EPIAS Transparency API	JSON	1 GB/yr	Hourly	Kafka Stream
FX & Agri Input Index	CBRT EVDS API	XML / JSON	<100 MB	Daily/Monthly	Airflow Batch

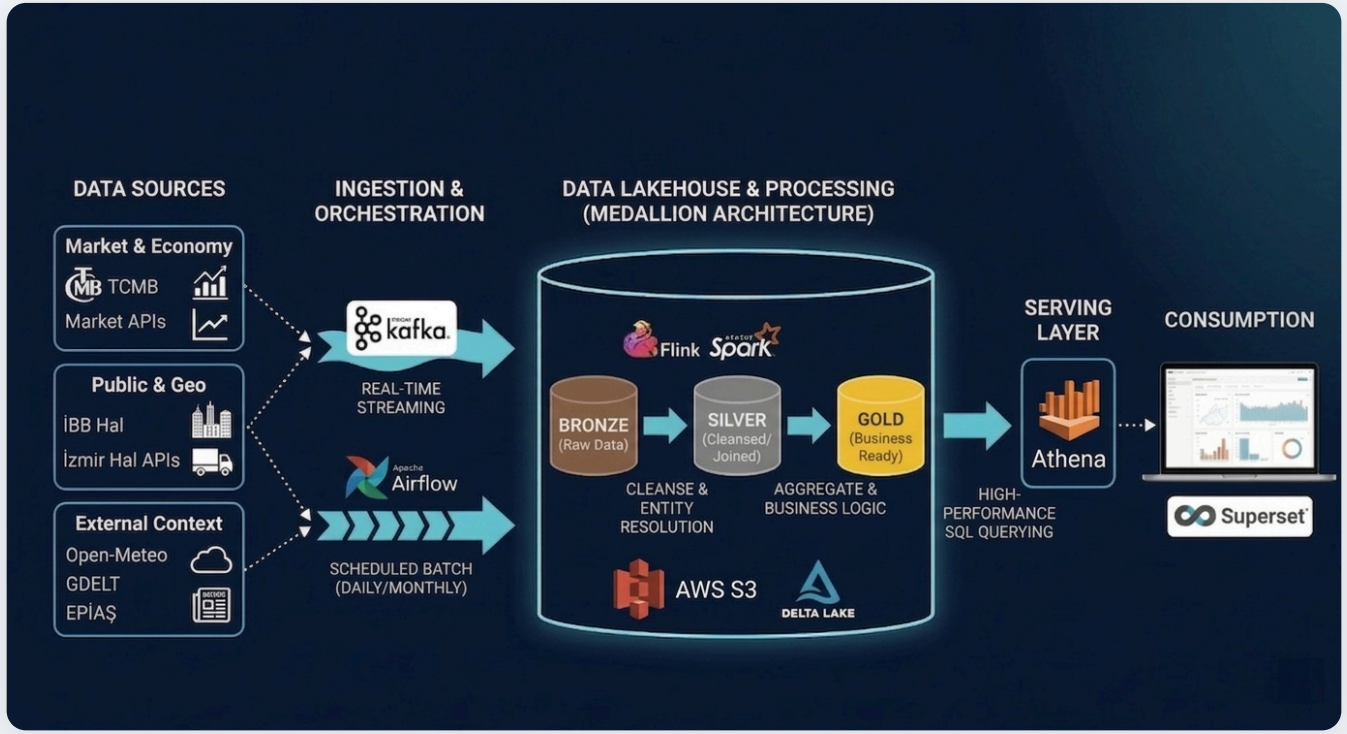
50–70 GB
INITIAL WITH 5–10 YR HISTORICAL DATA

→

200–500 GB
AT FULL 81-PROVINCE SCALE — TRUE BIG DATA

PIPELINE ARCHITECTURE

Data Journey: From Source to Insight



Real-time streams (Kafka) and scheduled batch jobs (Airflow) feed raw data into the **Bronze** layer. Apache Flink & Spark handle entity resolution and cleansing in the **Silver** layer. Aggregated margin and shock-propagation metrics are served from the **Gold** layer via Athena → Superset.

DATA ENGINEERING

Three Hard Problems We Solve

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CHALLENGE 1 · ENTITY RESOLUTION

Hall API: `"Domates Sofralık Sera"`
Retail slug: `"salkim-domates-1-kg"`

No shared ID exists between systems. **Solution:** TF-IDF + Cosine Similarity NLP pipeline via Spark MLlib matches products across schemas with no manual mapping.

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CHALLENGE 2 · UNIT STANDARDIZATION

Hall pricing: `Kasa / Çuval / Bağ`
Retail pricing: `Gram / Adet / Paket`

Solution: Custom Spark UDFs convert all units to **1 KG equivalent** before any price comparison or aggregation.

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CHALLENGE 3 · TEMPORAL & SPATIAL ALIGNMENT

GDELT: 15-min · EPIAS: hourly
Hall: daily · Tonnage: monthly

Solution: Delta Lake *As-of Join* aligns mismatched time granularities. *Spatial Join* maps each province's retail data to its nearest wholesale hall.

Stream Processing

Apache Flink for real-time entity resolution, unit conversion, and anomaly flagging on the incoming Kafka stream — before data hits Bronze.

Batch Transformation

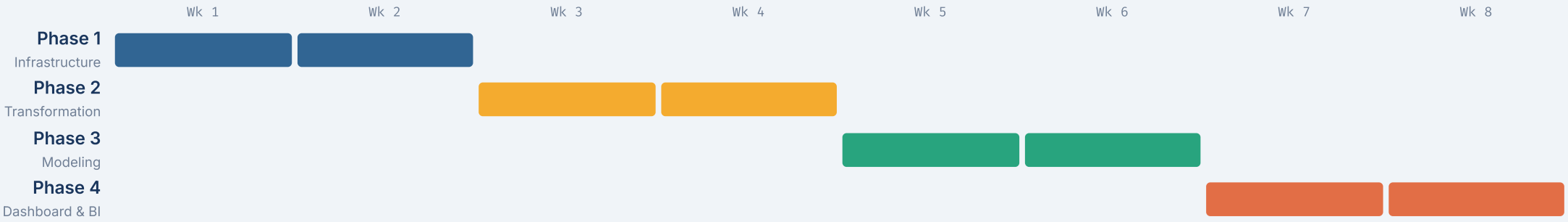
Spark Structured Streaming + Airflow-triggered jobs rebuild Silver and Gold layers nightly, incorporating all newly ingested batch data from halls and CBRT.

Anomaly Detection

Isolation Forest on the Silver stream flags when hall prices hold steady but retail price spikes ≥50% — distinguishing data errors from genuine speculative pricing events.

IMPLEMENTATION PLAN

8-Week Roadmap · 4 Phases



Phase 1
Weeks 1–2

Infrastructure & Ingestion — Kafka cluster, Airflow DAGs (IBB/Izmir Hall, CBRT), Bronze Delta Lake layer, Docker Compose

Phase 2
Weeks 3–4

Transformation & Entity Resolution — Spark/Flink Silver layer, TF-IDF NLP matcher, unit UDFs, as-of temporal join

Phase 3
Weeks 5–6

Modeling & Aggregation — Gold layer tables, Isolation Forest anomaly detection, spatial join (province ↔ hall), shock propagation index

Phase 4
Weeks 7–8

Dashboard & BI — Superset: Turkey Margin Map, Rockets & Feathers radar, GDELT event overlay, final reporting

BUSINESS & STRATEGIC VALUE

A Critical National Decision Support System

*Food price volatility is one of Turkey's largest structural economic problems. This engine turns it into **measurable, actionable, real-time intelligence**.*

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Regulators & Government

- Detect unjustified price spikes with **live evidence**
- Competition Authority enforcement data
- Ministry of Trade early-warning alerts
- Province-level speculation identification
- Evidence-based penalty mechanisms

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Consumers & Society

- Supply chain **transparency** dashboard
- Separate inflationary psychology from real cost
- Media-driven fear vs. genuine shortage
- District-level price fairness index
- Seasonal & weather shock education

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Economic Strategy

- Provinces with **logistics bottlenecks** flagged
- Retailers exploiting asymmetry identified
- Supply shock resilience scored per product
- Energy cost pass-through quantified
- Agricultural infrastructure investment signal

Long-term vision: An always-on national food price intelligence layer — the radar that regulators, researchers, and journalists can query in real time.

MEET THE TEAM

Our Team

HD

Hidayet Ersin Dursun

DATA SCIENTIST · DE

- Kafka cluster design & topic architecture
- Bronze Delta Lake ingestion layer
- Airflow DAG authoring (Hall APIs, CBRT)
- Docker Compose environment setup
- Rockets & Feathers statistical analysis

AZ

Abdullah Zengin

DATA ENGINEER · DS

- Apache Flink streaming transformation jobs
- Silver layer: null handling & unit UDFs
- Temporal as-of join & spatial join logic
- Isolation Forest anomaly detection pipeline
- Athena query layer optimisation

AY

Azmi Yağlı

DATA ENGINEER · PROJECT MANAGER

- Gold layer aggregate model design
- Shock propagation index computation
- TF-IDF + Cosine Similarity entity resolution
- Apache Superset dashboard development
- Real-time stream monitoring & alerting

By treating Turkey's food pricing opacity as a data engineering challenge, we transform a national black box into a transparent, queryable intelligence layer.

Thank you · Questions welcome